Guardian Interactive Security System

Final Project Report

Pajaka Lakshmin
Yongyan Liang
Briana Moyer
Parnika Nammi

TABLE OF CONTENTS:

Document Revision History

Revision Number	Revision Date	Description	Rationale
1	7/08/2024	Added: “System Overview”, “System Interfaces”, “System Purpose”	Initial Specs
2	8/07/2024	Updated: “Communications” Updated the table of contents.	Removed section instructions and clarified components using SPI.
3	8/30/2024	Updated: “Assumptions and Dependencies” “Project Description” “Problem Statement” “System Purpose” “System Context” “User Characteristics” “Operational Scenarios” “Bill of Materials (BoM)“	Fixed all initial thoughts and changes to finalized product
4	8/30/23	Updated: “System Capabilities and Constraints”, “Testing Strategy”	Changed capabilities and added new explanations for testing

Definitions, Acronyms, and Abbreviations

- **Biometric Data**: *Physiological or Behavioral Characteristics Used for Identification*

Biometric Data refers to unique physical or behavioral characteristics, such as facial recognition or voice patterns, that are used to identify and authenticate individuals.

- **GISS**: *Guardian Interactive Security System*

  The Guardian Interactive Security System is an advanced security solution designed to monitor and protect environments through a combination of embedded systems, real-time surveillance, and intelligent automation.

- **GPIO**: *General Purpose Input/Output*

  A generic pin on an integrated circuit (IC) whose behavior (input or output) can be controlled by the user at runtime. GPIO pins are used for interfacing with various external devices.

- **Micro Servo SG90**: *A Small Actuator for Precision Control*

  The Micro Servo SG90 is a compact servo motor commonly used in embedded systems for precise control of angular or linear position, velocity, and acceleration.

- **PWM**: *Pulse Width Modulation*

  A method used in electrical engineering to control the amount of power delivered to a load without losing efficiency. It's commonly used in controlling the speed of motors, brightness of LEDs, etc.

- **RGB LED**: *Red-Green-Blue Light-Emitting Diode*

  An RGB LED is a type of LED that combines red, green, and blue light-emitting diodes in a single package. By adjusting the intensity of each diode, a wide range of colors can be produced. RGB LEDs are commonly used in displays, indicators, and lighting systems to create various color effects.

- **UI**: *User Interface*

  The User Interface (UI) refers to the visual and interactive elements of a system or application that users engage with. It includes components such as buttons, menus, and icons that allow users to interact with the system and control its functionalities.

- **USB**: *Universal Serial Bus*

  USB (Universal Serial Bus) is an industry-standard interface that allows communication between devices and a host controller, such as a raspberry pi. It supports the connection of peripherals like cameras, microphones, storage devices, and more, facilitating data transfer and power supply.

Assumptions and Dependencies

This project assumes that the user will only be using this system for one room as there is no system to allow communication between multiple cameras and systems. The scope of the camera and motion sensors are also limited to one room. We also assume that the system is always plugged into a source and that the user will disarm the system when bringing unauthorized guests to avoid unwanted alarms.

This project will be dependent on a constant power source. The Guardian should operate from a standard 15 amp outlet. It is also dependent on the fact that it is mounted in an area where faces are recognizable and, if decided, the camera has the ability to rotate.

Project Description

The Guardian Interactive Security System (GISS) is an interactive monitoring system that uses advanced voice and facial recognition features. Its purpose is to monitor a room and sound an alarm when unauthorized users are detected. It consists of three major components: an interactive LCD screen that functions as the UI, a buzzer that will sound when the intruder is recognized, and a webcam that will monitor. Our LCD screen provides an interface where users can add new users, remove existing users, and arm/disarm the system. When adding a new user, the LCD guides the user through setting up their 3-step verification: facial recognition, voice recognition, and pin code authentication. Upon setup, the admin registers first with their biometric data and a name the system will use to reference them. They are granted sole access to removing and adding users from the system. To operate, the admin may use a saved voice command to arm the system and the camera will begin to monitor until an authorized user disarms it by entering a pin code. When an unrecognized user is detected, the alarm sounds until the disarm procedure is completed.

Our inspiration for this project comes from our combined interest in machine learning and security. This project incorporates embedded systems using a camera, microphone, buzzer, raspberry pi, touchscreen, and an RGB led. GISS is also equipped with pretrained models for voice features and facial recognition.

The admin of the system is granted primary access to priority security features such as voice arming and managing user permissions and accessibility as needed. The system is capable of securely storing each user's permission levels, security passcodes and biometric data. These features are aimed at providing the admin a convenient and efficient interaction with their security system. Permissions cover a variety of scenarios in which the user may come across. For example, they may want to allow their toddler to roam freely without setting off alarms but not necessarily capable of arming the system. This is one of many ways that the admin may customize their security experience.

GISS is also equipped with three methods of alerts. A blue light indicates the system is disarmed, and green light for armed, and red when there has been a security breach. Additionally, an alarm will sound and the admin will be notified via email of a potential intruder.

Using modern advances in machine learning, this system improves security using facial and voice recognition software. The facial recognition feature identifies authorized versus unauthorized users. If a person is detected and not registered as a user, a buzzer (or alarm) will sound, signifying that there is an unknown person. This is accomplished by pre trained and tested software libraries such as OpenCV and PicoVoice.

The voice recognition system will also be able to recognize authorized, or registered, users, as well as identify unique voice passcodes as initially set up by the individual user. They may use phrases like “Arm System” or “Lock the house up”. Once this happens, the system checks the users permissions in real time. If the user is authorized, GISS will switch to the armed state. Otherwise, if the user does not meet voice activation permission levels, the system will remain unarmed.

Voice verification also provides additional security in the event that a person attempts to bypass facial recognition, adding an extra level of protection. Persons detected who fail facial and voice security measures will trigger the alarm while in the armed state. In the case where a user meets both these security requirements, a pin code is required to fully disarm the system. The pin code is kept secure on the admins local device, allowing each user to create their own unique pin code upon registration. When entering a pin code, the user is allowed three attempts. Each failed attempt will prompt a warning message on the console. After the third failed attempt, the alarm will be triggered. This is how GISS provides 3-step verification for added protection.

Problem Statement

Our inspiration for this project comes from our combined interest in machine learning and security. This project incorporates embedded systems using a camera, mic, buzzer, raspberry pi, touchscreen, and lights. We will also be using libraries to recognize faces and voices.

The problem that we are aiming to solve is the security of a room to avoid intruders in unwanted areas. The Guardian Interactive Security System is meeting the need for a secure room monitoring system that detects and responds to unauthorized access for people who want to keep important things safe.The GISS includes facial recognition, voice recognition, and password authentication to enhance security which ensures that only authorized users are able to come into the room with the system. It also provides a good method of monitoring and securing rooms. The system's user interactive features include adding or removing users and arming/disarming making it easy to use while keeping security.

System Purpose

System Name: Guardian Interactive Security System

Application

The Guardian Interactive Security System, or GISS, is a monitoring device that provides enhanced security through advanced recognition and alert mechanisms. It uses facial and voice recognition with pin code authentication to provide a comprehensive and user-friendly security experience.

Top-Level Benefits, Objectives, and Goals

1. **Enhanced Security**
   - **Intruder Detection:** The Guardian continuously scans a room with a 360-degree rotating camera. It sounds an alarm when the camera detects an unregistered user.
   - **Multi-Factor Authentication:** The Guardian implements facial recognition, voice recognition and pin code authentication to arm/disarm the security system which reduces the risk of unauthorized security breaches.
2. **User-Friendly Interface**
   - **Interactive Touchscreen:** The Guardian includes an attached touchscreen, allowing the user to add a new user, remove an existing user, and arm/disarm the Guardian's security system.
   - **Voice Commands:** An authorized user's voice arms or disarms the system for hands free operation.
3. **Administrative Control**
   - **User Tracking:** The Guardian maintains a record of user permissions.
   - **Permission Management:** The admin has full control over individual user permissions, including adding/removing users, and setting permissions for other users to arm/disarm the Guardian's security system.
4. **Reliable**
   - **Alerts:** A buzzer (alarm) activates when an unrecognized person is detected in the room. We will also use motion sensors to ensure that no one can surpass the system. This alerts nearby users or personnel to the potential security breach.
   - **Advanced technologies:** The Guardian implements libraries to accurately identify registered users vs unregistered.

System Context

Our system can be used in any places that need security monitoring to keep the place safe from unwanted intruders. Our system provides monitoring of the room and security alarm if unrecognized people enter the room. Our system can be used to keep homes, companies, banks, etc safe from unwanted people from entering their homes and companies plus keeping a 24 hour monitoring system to detect any suspicious activity.

Our system includes different components with a goal to monitor the room and facial recognition with a camera, microphone for voice recognition and voice commands, buzzers and lights for the alarm system, an interactive lcd touch screen for user interaction and pin code verification. Our system utilizes libraries to learn designated faces for facial recognition, designated voices for specific voice commands to ensure certain actions are only done through specific users of the system.

All significant interfaces crossing the system's boundaries include the camera, microphone, buzzer, lcd touchscreen, and raspberry pi. The raspberry pi is our main processing unit running our three layers of verification and the interfaces are the camera, microphone, and LCD touchscreen. The camera is the main verification for the security system with live video monitoring of the room and facial recognition. The interface is the usb connection or pins to raspberry pi. The microphone is another verification for the system that captures specific voice commands and voice recognition. The interface is the usb connection or pins to raspberry pi. The buzzer is the alarm system for the security system that will be triggered if an unrecognized person comes into the room. The interface is the usb connection or pins to raspberry pi. The lcd touch screen is an interactive system for the user to interact with that can set up accounts for the users and enter pin codes for necessary actions to be done. The interface is the hdmi connection or pins to raspberry pi. We also have motion sensors that are add ons to help the camera detect movement and the interface is the pins to raspberry pi.

User Characteristics

The users of our system are anyone who wants to make a room more secure. Our system will monitor the room and sound an alarm when an unrecognized person enters. There are two different types of users for the system. One type of user is regular users. Regular users are split into authorized users and regular unauthorized users. The regular users won't sound the system alarm because they will be recognized. The users will go through the lcd touch screen which will guide the user through setting up three verification methods: facial recognition, voice recognition, and pin code authentication plus a name for the account.

The authorized user can do more than regular users. The authorized user can give the voice command to arm the system and the camera will monitor until the deactivation command is spoken and the pin code is entered.

The other type of user is admin users. The admin of the system has the ability to arm or disarm the system at any time using all 3 authentication methods and edit the permissions of the other users. The admin of the system can remove users from the system. Other users will not be able to arm or disarm the system unless the admin gives them permission to do so which means that they are going from regular users to authorized users with this permission.

The functions of the admin users is that they have full control over the whole system because they have the biggest authority with control over the system. The functions of the regular users is that they won't trigger the alarm and the functions of the authorized users is that they have permission from the admin to have access to disarm and arm the system. The location is where the system is placed to monitor and for the security of the certain buildings and places. The type of device the users uses is the lcd touch screen plus both the video camera and voice microphone for interactions and allowing access.

Operational Scenarios

Andrew has newly purchased his Guardian Security System and needs to set up his face along with his family's. He turns on the system and clicks "set up profile". Here he is able to set up his face ID, voice ID, and pin code. He first inputs his name (first and last) and then since there is not a main user set up yet, the system will ask him to verify that he is going to be a main user. This will set his profile to have certain permissions. It will then scan his face to set up the face ID. To set up the voice function, it will ask him to choose the phrases he would like to set up. After speaking clearly, the system will play them back to ensure that it sounds good. After confirming, it will ask to set up a pin code. He will enter the pin code twice and the system will make sure that both instances match. He has now completed setting up the main authorized user profile.

Andrew will then be able to set up his family's profiles following the same steps, except he will choose which permissions he wants to give to each of his family members. For example, he could set it up such that only his partner is the only other person who can arm and disarm the system but his children will not be able to.

His child, Beth, attempts to arm the system using a voice command, however, the system will reject it as her profile is not set with the permissions needed to arm the system. After Andrew arms the system, Beth is able to disarm the system but not with her voice, only with her special pin code.

Bill of Materials (BoM)

Only the final materials that went into the final product. Including any items that were already owned at the beginning of the quarter.

Part #	**Description**	**Price per unit**	**Quantity**	**Total price**
Camera	for Raspberry Pi 4 B 3 B+ Camera Module Automatic IR-Cut	$77.99	1	$77.99

	Switching Day/Night Vision Video Module Adjustable Focus 5MP OV5647 Sensor 1080p HD Webcam for Raspberry Pi 2/3 Model B Model A A+			
Touchscreen	3.5 Inch 480x320 Touch Screen TFT LCD SPI Display Panel for Raspberry Pi A, B, A+, B+, 2B, 3B, 3B+,4B,5	$17.99	1	$17.99
Raspberry Pi	Raspberry Pi 4 Computer Model B 8GB Single Board Computer Suitable for Building Mini PC/Smart Robot/Game Console/Workstation/ Media Center/Etc.	$77.90	1	$77.90
Motion Sensors	5Pcs HC-SR501 PIR Motion Sensor Module + 2Pcs Exclusive housing for Arduino Raspberry Pi Toys, Lamps, Electronic Equipment	$9.99	1	$9.99
Mic	SunFounder USB 2.0 Mini Microphone for Raspberry Pi 4 Model B, Module 3B+, 3B 2 Module B & RPi 1 Model B+/B Laptop Desktop PCs Skype VOIP Voice Recognition Software	$7.99	1	$7.99
Micro SD Card	SanDisk 32GB (Pack of 2) Ultra microSDHC UHS-I Memory Card (2x32GB) with Adapter - SDSQUA4-032G-GN6MT [New Version]	$12.87	1	$12.87

SD Card Adapter	USB3.0 Micro SD Card Reader, 5Gbps 2-in-1 SD Card Reader to USB Adapter, Wansurs Memory Card Reader for SDXC, SDHC, MMC, RS-MMC, Micro SDXC, Micro SD, Micro SDHC and UHS-I Cards (1Pack Black)	$4.99	1	$4.99
TOTAL	-	-	-	**$287.62**

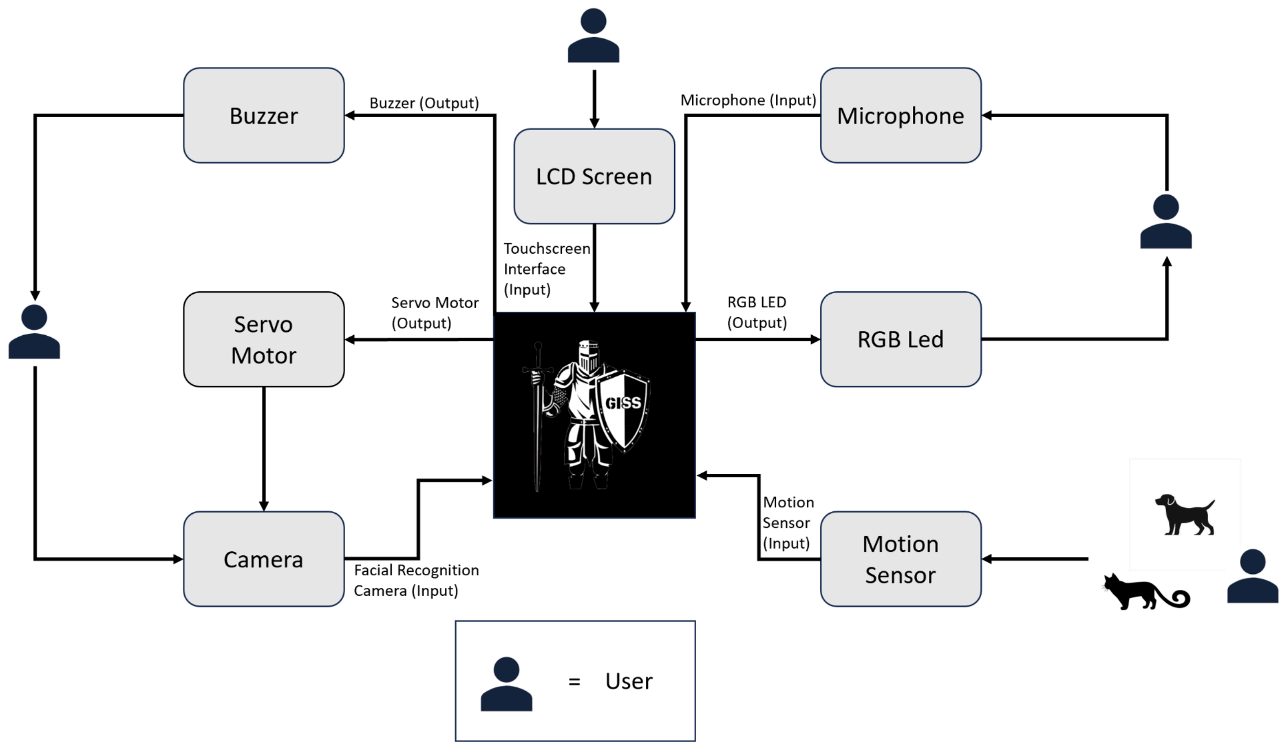
Engineering Standards

Description	**Standard**	**Governing body**
Floating point	IEEE 754-2008	IEEE
Verification and Validiation	IEEE Std 1012-2016	IEEE
Facial Recognition	ISO 19794-5	ISO
Audio Recognition	AES3	AES
Passwords	NIST- (SP) 800-series	NIST
Motion Sensors	NEMA	NEMA

System Overview

The system consists of the following major components:

- The buzzer serves as the alarm that sounds during a security breach.
- The LCD display allows the user to set up an account, add/remove users, change permission settings and arm/disarm the system.
- The servo motor rotates the camera.
- The USB Lavalier microphone allows voice input to be used by the software for processing voice biometric data.
- The motion sensors detect nearby movement in a room.
- The RGB led indicates the status of the system.

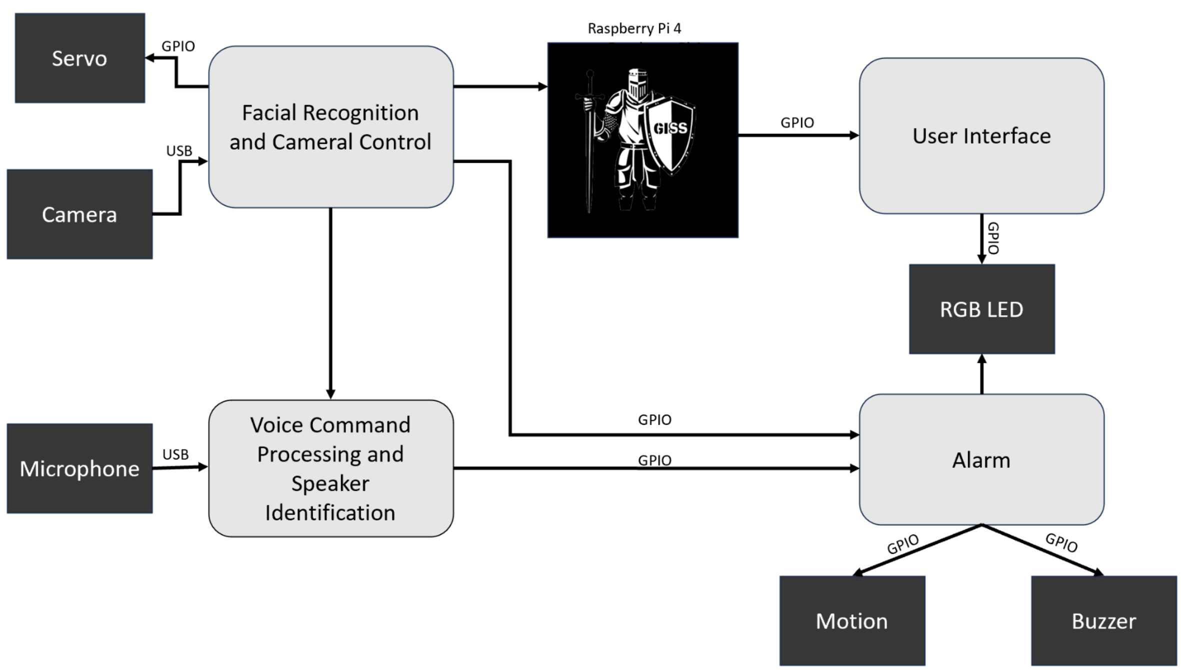


Input/Output

Name	I/O	Description	Use
Camera	Input	Captures live video to monitor the room and perform facial recognition to identify registered users.	Triggers an alarm if an unrecognized user is detected.
Microphone	Input	Receives voice commands and captures voice biometric data from the user.	Used for voice recognition to arm/disarm the system, identify speakers, and register new users.
Motion Sensors	Input	Detects movement within the room.	Activates pedestrian detection to determine if the detected movement is a person.
LCD Touch Screen	Input	Allows users to interact with the system by adding/removing users, entering pin codes, and arming/disarming.	Serves as the primary user interface for setup and control.
Buzzer	Output	Sounds an alarm when the system detects an unrecognized user or incorrect pin codes are entered multiple times.	Provides an auditory alert to signal potential security breaches.

RGB LED	Output	Provides visual status indicators for the system.	Displays a green light when the system is armed, blue light when disarmed, and red light when the alarm is sounding.
Servo Motor	Output	Rotates the camera to provide a 360-degree view of the room.	Adjusts the camera's angle based on detected motion to ensure comprehensive monitoring.

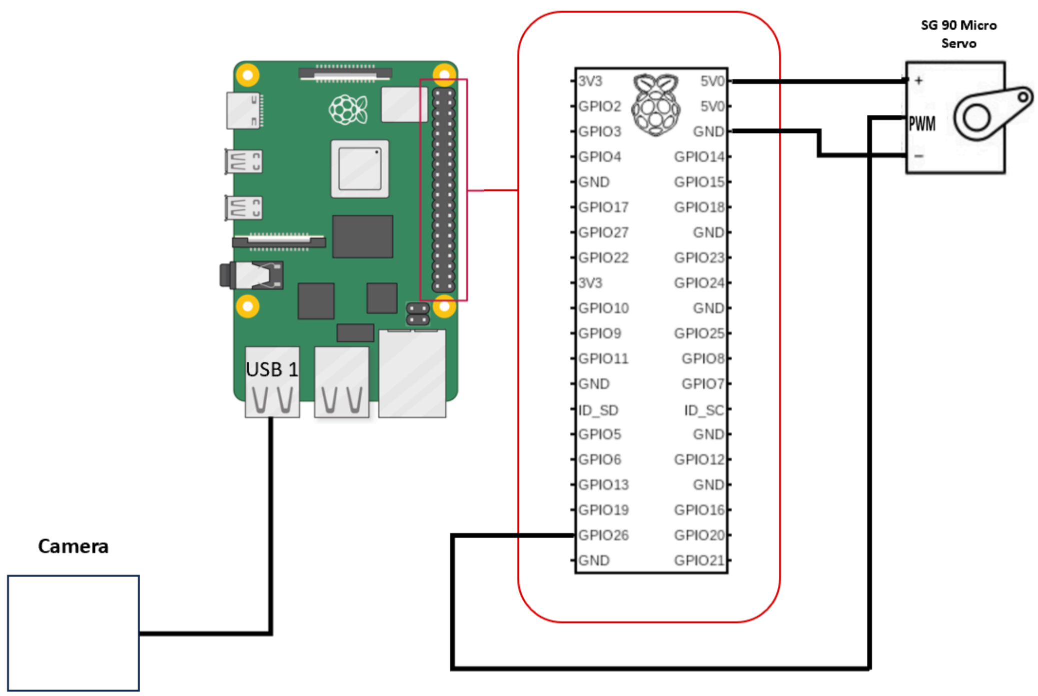
Subsystems



Facial Recognition and Camera Control Subsystem

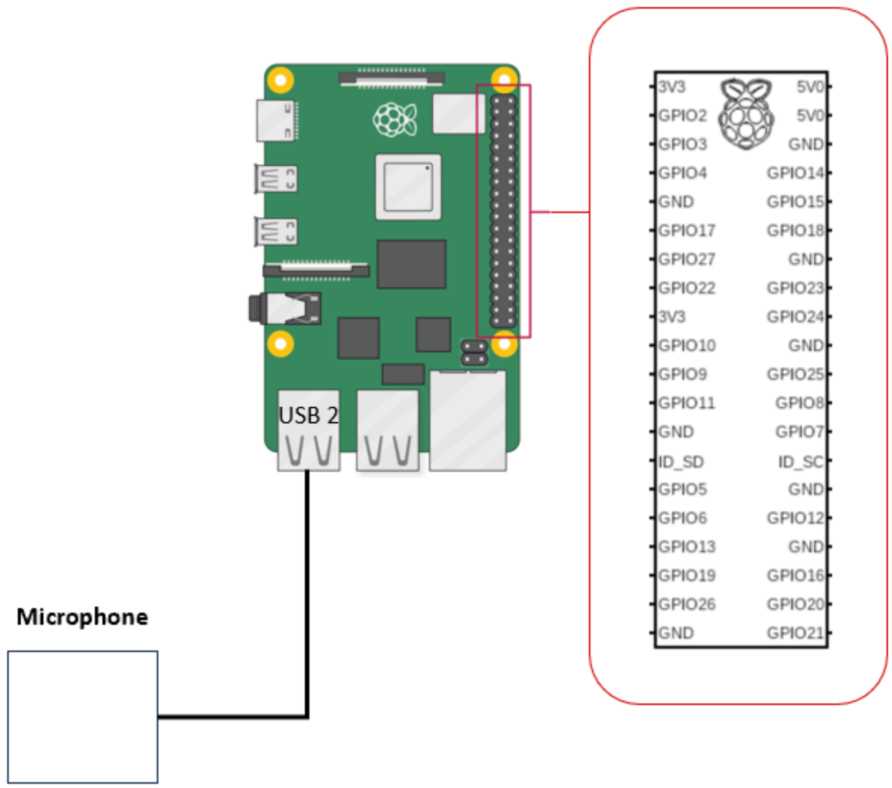
This subsystem is responsible for video/image capture, performing facial recognition, and controlling the camera's rotation to ensure comprehensive monitoring. The camera feed is analyzed in real-time to detect and identify users. If an unrecognized user is detected, the system triggers an alarm.

Name	I/O/Comm	Description	Use
Camera	Input	Captures live video for facial recognition.	Monitors the room and identifies users based on facial profiles.
Servo Motor	Output	Rotates the camera to cover a 360-degree area.	Ensures comprehensive monitoring by adjusting the camera angle based on detected motion.
GPIO	Comm	Communication between Raspberry Pi and camera.	Controls the servo motor to adjust the camera angle.
USB	Comm	Communication between Raspberry Pi and camera.	Transfers video data from the camera to the Raspberry Pi for processing.



Voice Command Processing and Speaker Identification Subsystem

This subsystem handles voice recognition, wake word detection, and speaker identification to allow users to interact with GISS. The microphone captures the voice input which is processed to identify the speaker and execute commands to arm/disarm the system or trigger the alarm.

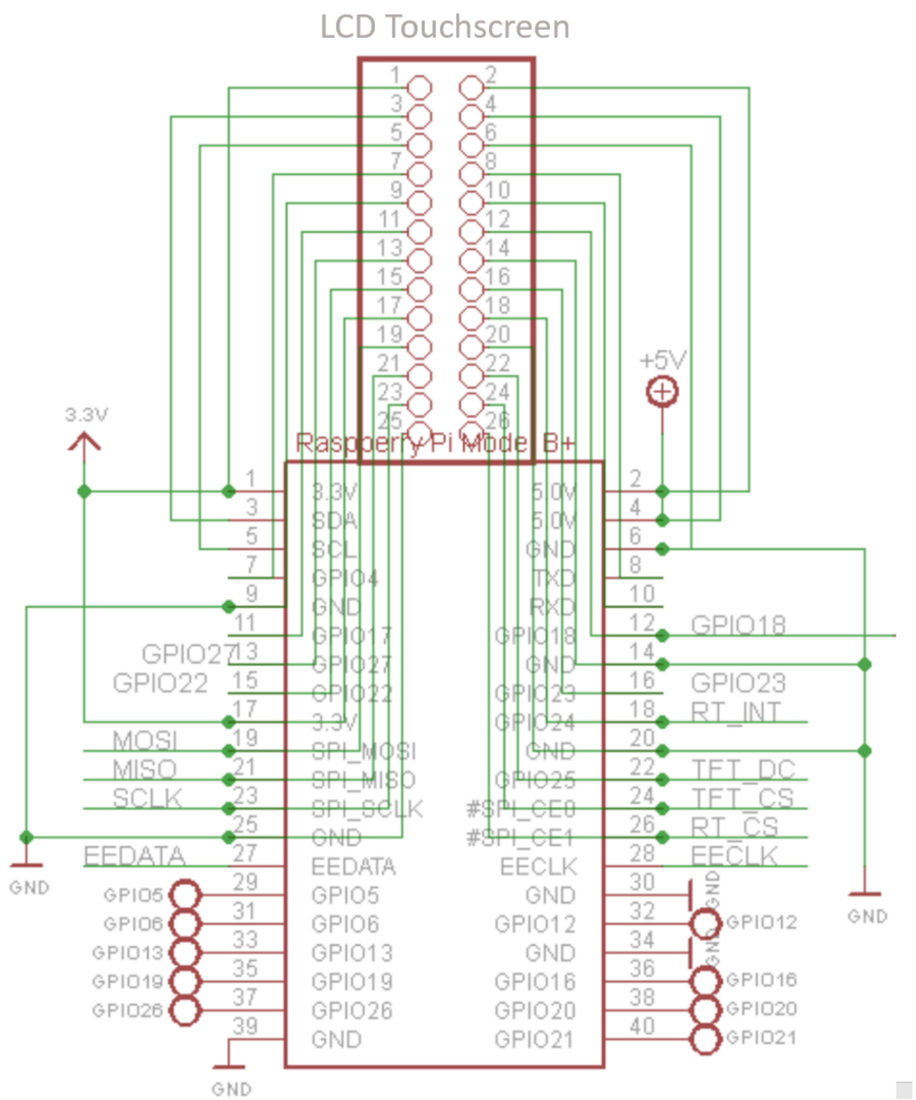


Name	I/O/Comm	Description	Use
Microphone	Input	Captures voice commands and speaker biometrics.	Enables voice-based interaction with the system, including arming/disarming the system or triggering the alarm buzzer.
GPIO	Comm	Communication between Raspberry Pi and buzzer.	Transfers audio signals for output to the buzzer.
USB	Comm	Communication between Raspberry	Transfers voice data from the microphone

		Pi and microphone.	to the Raspberry Pi for processing.

User Interface Subsystem

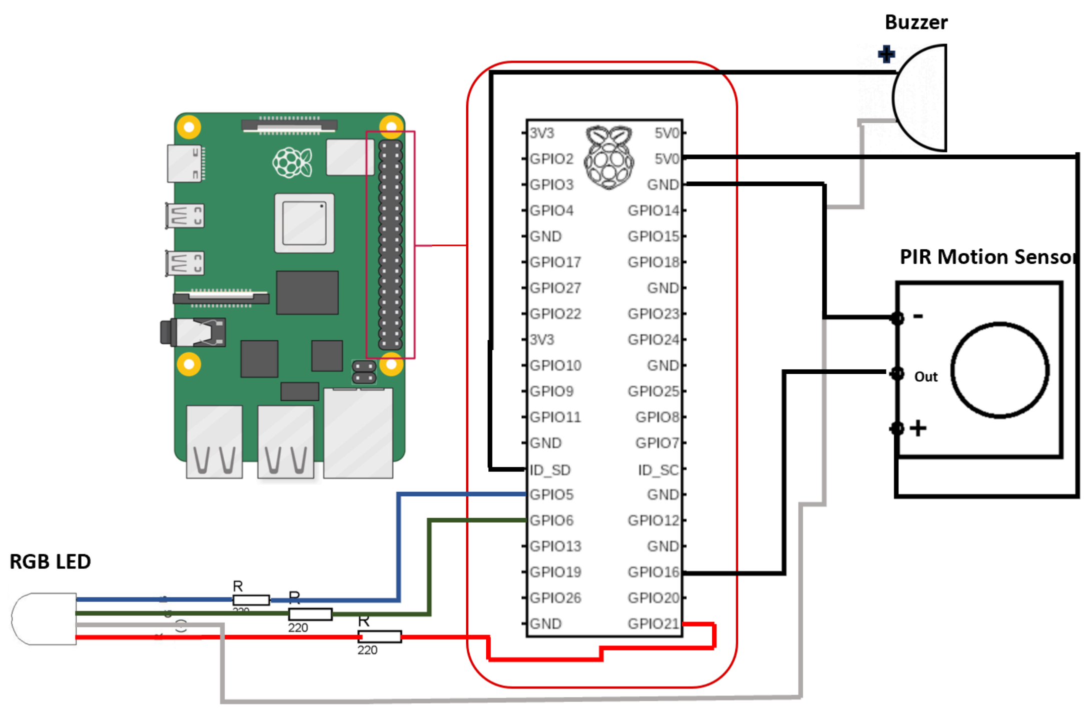
This subsystem contains the UI components within the LCD touch screen. It provides an interactive platform for users to manage the system and user profiles.



Name	I/O/Comm	Description	Use
LCD Touch Screen	Input	Allows user interaction.	Facilitates user control over system settings, including arming/disarming and user management.
GPIO	Comm	Communication between Raspberry Pi and touchscreen/RGB LED.	Transfers data for display and status indication.

Alarm Subsystem

This subsystem alerts users to unauthorized access or security breaches. It controls the buzzer for audible alerts, the RGB LED for visual alerts, and the motion sensor for detecting movement. The motion sensor triggers further action such as activating the camera for pedestrian detection.



Name	I/O/Comm	Description	Use
Motion Sensor	Input	Detects movement within the room.	Triggers the system to activate the camera and initiate pedestrian detection when motion is detected.
Buzzer	Output	Provides auditory feedback.	Communicates system status and feedback to the user.
RGB LED	Output	Displays system status.	Visually indicates whether the system is armed, disarmed, or in an alert state.
GPIO	Comm	Communication between Raspberry Pi and motion sensor.	Transfers data regarding detected motion to the Raspberry Pi for processing.

System Capabilities, Conditions, and Constraints

Physical

Construction
The entire system is stationary and will be no larger than 1ft x 1ft x 1ft. It is meant to be kept on a medium high surface like a shelf. It is easy to transport and once it has been set up it can be moved to its permanent station. The entire system will consist of a mounted camera, a touchscreen, and speaker plus motion sensors placed in different areas of the room to detect movement.

Adaptability
This system can be adapted to be more secure. This can include adding more security features like a fingerprint scanner, another camera with motion detection, or an app that can be used to monitor remotely. We could also add extra data security to ensure that people cannot bypass the security system by hacking into it.

Environmental Conditions
The Guardian is an indoor security system and should not be placed outside of a home or office. The Guardian has not been tested to withstand intense heat, wind, or rain. It is also not water-proof so it should be placed away from any water.

Communications

There are no requirements for the frequency of communication

Storage
The data will be stored forever because the data that we are storing will be the user data with their facial recognition data, voice recognition data and their set pin codes. This user data will be stored forever unless the user decides to delete their account or the admin decides to delete a certain user from the system/database. The requirements for associated archival storage is that the data will be stored encrypted to ensure the security of the system. The most important requirement is keeping the user data safe and secure so that our important user data will not be compromised.

There are two types of data: user data and the system data. The system data will store logs and records of access, when the alarm is triggered, administrative actions of arming and disarming the system to keep track of suspicious activity. This data will be stored for about a year to monitor suspicious activity and patterns and ensure safety. The requirements for associated

archival storage is that this data will be stored encrypted as well to keep the data safe and ensure security.
Distribution

Our system is kind of distributed with the components of our system. Our system is distributed among three big components which is the camera for facial recognition, the microphone for voice recognition, and the main raspberry pi system with the lcd touch screen for the pin code password. These three components are the three big verification steps of our security system. The main component is the camera component for facial recognition. The other components are the microphone component and the lcd touchscreen component for the voice recognition and pin code access for additional layers of security plus motion sensors to assist the camera to detect movement.

Sensors

We will need sensors for the camera and the microphone. The first sensor will be the camera sensor that we need for the camera to monitor the surroundings and be able to allow the user to access the facial recognition feature. This sensor needs to take video or images for facial recognition. The precision requirements for this sensor is that the camera sensor should have high resolution to detect people's faces clearly in different lighting conditions such as day and night.

The second sensor will be the microphone audio sensor that we need for the microphone to detect audio voice to allow the user to access the voice recognition feature. This sensor needs to take in audio voice commands for voice recognition and to disarm or arm the system. The precision requirements for this sensor is that the audio sensor should be highly sensitive to capture people's voices no matter the volume that they speak at. Another requirement is that the audio sensor should be able to distinguish between human voices and other surrounding sounds that are not important for the system to detect to get clear audible human voices with no background noises interrupting with the system voice recognition.

The third sensor we need is a motion movement sensor to detect movement to help the camera know where movement is happening. This motion sensor needs to detect the room and tell the camera if there is any movement to avoid people facing away from the camera to get away with facial recognition. The precision requirements for this sensor is that the motion sensor should be highly sensitive to detect any slight movement from a person in the room. Another requirement is the range of detecting the motion movement because the motion sensor should be able to cover the whole room so no corner is left undetected.

System Performance Characteristics

A critical performance condition is the accuracy of facial recognition. The associated capabilities is that the system should accurately recognize the faces of registered users and unrecognized users. Another critical performance condition is the accuracy of voice recognition. The associated capabilities is that the system should accurately recognize the voices of authorized registered users and unrecognized users.

Another critical performance condition is the live time monitoring of the camera. The associated capabilities is that the system should be constantly monitoring the room with the camera. Another critical performance condition is the alarm system. The associated capabilities is that the system should trigger the alarm system immediately when an unrecognized person is detected to ensure the safety of the place.

Another critical performance condition is the user interface. The user should be able to operate the lcd touch screen and get their actions done quickly. The associated capabilities is that the system should trigger the user interface actions immediately when a user is trying to create their account or do any other actions. Another critical performance condition is the security of user data. The associated capabilities is that the system should keep user data encrypted and secure.

System Interfaces

Components and Their Interfaces

1. Interactive Touch Screen
    - Interface Characteristics:
        - Touchscreen input for user interaction.
        - Display output for showing setup options, status messages for the user.
    - Communication Protocols:
        - Communicates with the Raspberry Pi
    - External Capabilities:
        - Allows users to add or remove users, arm/disarm the system with a pin code, and manage settings.
    - Constraints:
        - Requires a clean and responsive touchscreen environment.
        - Requires proper power supply and configuration to the Raspberry Pi.
2. Raspberry Pi
    - Interface Characteristics:
        - Microcontroller for handling inputs from the camera, microphone, and touchscreen.
        - Outputs control signals to the buzzer and touchscreen.
    - Communication Protocols:

        - Interfaces with the buzzer, microphone, camera, light indicator, touchscreen and motor.
    - External Capabilities:
        - Executes machine learning algorithms for facial and voice recognition.
        - Manages user permissions and system status.
    - Constraints:
        - Requires adequate power supply and network connectivity for software and communication.
3. Camera
    - Interface Characteristics:
        - Video input to the Raspberry Pi for detection.
        - Motor control interface for 360-degree rotation.
    - Communication Protocols:
        - Communicates with the Raspberry Pi.
    - External Capabilities:
        - Captures images and video for facial recognition.
    - Constraints:
        - Requires a stable mounting and rotation mechanism.
4. Microphone
    - Interface Characteristics:
        - Audio input to the Raspberry Pi for voice recognition and voice commands.
    - Communication Protocols:
        - Communicates with the Raspberry Pi.
    - External Capabilities:
        - Captures voice commands for arming/disarming the system.
    - Constraints:
        - Requires a quiet environment for accurate voice recognition.
5. Buzzer
    - Interface Characteristics:
        - Audio output to signal alarm.
    - Communication Protocols:
        - Controlled by the Raspberry Pi.
    - External Capabilities:
        - Sounds an alarm when an unrecognized person is detected.
    - Constraints:
        - Requires proper power supply and configuration to the Raspberry Pi.
6. User Inputs and Outputs
    - Interface Characteristics:
        - User inputs include touchscreen taps, facial recognition data to camera, voice commands, and PIN entries.
        - Outputs include visual feedback on the touchscreen, audio alerts from the buzzer, and status lights.
    - Communication Protocols:

        - User inputs are processed via the touchscreen, microphone, and camera.
    - External Capabilities:
        - Users can interact with the system to manage security settings and respond to alerts.
    - Constraints:
        - Requires clear user instructions and responsive interface elements.
7. SD Card
    - Interface Characteristics:
        - Memory
    - Communication Protocols:
        - Communicates with the Raspberry Pi
    - External Capabilities:
        - Allows users to perform any action and stores any relevant details that should be remembered such as user info.
    - Constraints:
        - might run out of memory
8. Motor for Raspberry PI
    - Interface Characteristics:
        - Moves the camera around
    - Communication Protocols:
        - Communicates with the Raspberry Pi
    - External Capabilities:
        - Allows camera restricted range of motion determined by the Raspberry Pi
    - Constraints:
        - might be too slow or fast causing image quality to be unclear
        - might miss movement or get distracted with multiple movements
9. Camera Mount
    - Interface Characteristics:
        - stand for stability
    - Communication Protocols:
        - n/a
    - External Capabilities:
        - Is used for motion.
    - Constraints:
        - Gets in the way of the motor
10. Motion Sensors
    - Interface Characteristics:
        - n/a
    - Communication Protocols:
        - Communicates with the camera and Raspberry Pi
    - External Capabilities:
        - Detects motion
    - Constraints:
        - Might take time to signal to the camera

Testing Strategy

Milestone 1 Components

1.1.1 Menu Initialization

- This test ensures that when the device is powered on the accurate menu is displayed for the user to set up the admin user profile
- Inputs: power button
- Output: Menu screen
- Test type: performance
- PASSED

1.1.2 User Profile Name

- This test ensures that the program saves the profile name correctly
- Input: User inputting name
- Output: Program saves name
- Test Type: performance
- PASSED

1.1.3 Admin Password

- This test ensures that the program saves the admin's password correctly
- Input: Admin inputting password
- Output: Program saves password
- Test Type: performance
- PASSED

1.1.4 Admin accessing User Permissions

- This test ensures that the program only allows the admin to be able to change a user's permissions and tests that the permissions actually change
- Input: Admin changing user permissions + test user stands in front of the system, ready to test
- Output: The permission changes are reflect with the test user
- Test Type: performance
- PASSED

1.1.5 Facial Recognition - Identify Unique Users

- The test verifies the facial recognition library can accurately identify different users.
- Input: Two users

- Output: User 1 and User 2 are both identified on screen.
- Test Type: Performance.
- PASSED

1.1.6 Facial Recognition - Identify Pets

- This test verifies that the facial recognition library performs proper identification of pets vs humans.
- Input: Human, dog, and cat feedback to camera.
- Output: The live camera shows a square around the human with a “human” label, and doesn’t show anything for pets.
- Test Type: Performance.
- PASSED

1.1.7 Voice Recognition - Unique User Identification

- This test verifies that the voice recognition software identifies unique voices.
- Input: User 1 speech (e.g. User Briana says ”Hello, how are you?” into the microphone)
- Output. “Briana’s voice identified” prints to terminal.
- PASSED

1.1.8 Facial Recognition - Identify Objects

- This test verifies that the facial recognition library performs proper identification of objects vs humans.
- Input: Pillow, cup, and charger feedback to camera.
- Output: The live camera shows a square around the human with a “human” label, and doesn’t recognize camera.
- Test Type: Performance.
- PASSED

Milestone 2 Components

2.1.1 Microphone-background noises

- This test ensures that the voice recognition can differentiate between human voices and background noises
- Inputs: human voice plus other background noises
- Expected output: the correct action based on the voice commands
- Test type: performance
- PASSED

2.1.2 Microphone-sensitivity

- This test ensures that the voice recognition can senses human voices regardless of volume
- Inputs: human voice at a distance
- Expected output: user voice recognized within 15 feet
- Test type: performance
- PASSED

2.1.3 Camera-quality

- This test ensures that the camera captures clear footage under different lighting conditions
- Inputs: human face in different lighting background
- Expected output: a clear image for facial recognition
- Test type: performance
- PASSED

2.1.4 Motion sensors-functionality

- This test ensures that the motion sensor actually detects a moving person
- Inputs: person walking
- Expected output: motion sensor detects movement
- Test type: performance
- PASSED

2.1.5 Motion sensors-data storage

- This test ensures that the motion sensor actually detects a moving person and activates the buzzer
- Inputs: person walking
- Expected output: motion sensor detects movement and communicates with buzzer if facial does not work
- Test type: performance
- PASSED

2.1.6 Microphone - Speech-to-Text Audio Quality

- This test verifies that a human voice can be picked up with various background noises.
- Input: Human speech with various ambient sounds
- Output: Human speech prints to terminal.
- Test type: Usability.
- PASSED

2.1.7 Voice Recognition - Unique Phrase

- This test ensures that the voice recognition software can identify unique phrases.
- Input: User speaks unique phrase (e.g. “Hold down the fort”)
- Output: “Voice passcode accepted, arming the Guardian” prints to terminal.
- Test Type: Performance.
- PASSED

2.1.8 Voice Recognition - Non-Unique Phrase

- This test ensures that the voice recognition software can identify non-unique phrases.
- Input: User speaks non-unique phrase (e.g. “Turn on the system”)
- Output: “Voice passcode denied” prints to terminal
- Test type: Performance
- PASSED

2.1.9 Voice Recognition - Enroll Multiple Users

- This test ensures that the voice library can save multiple users.
- Input: User runs the program, enters their name into the terminal and performs a voice test.
- Output: Terminal prints “(User’s name) was added successfully
- Test Type: Usability
- PASSED

2.1.10 Voice Recognition - Identify Multiple Speakers

- This test ensures that the voice library can identify multiple users.
- Input: User1 speaks into the microphone, followed by User2 and User3.
- Output: Terminal prints “User1 is speaking. User2 is speaking. User3 is speaking.”
- Test Type: Usability
- PASSED

2.1.11 Voice Recognition - Recognize Unauthorized Speakers

- This test ensures that the voice library can identify speakers who have not been registered and are not authorized.
- Input: Unregistered user speaks into the microphone.
- Output: Terminal prints “User not recognized.”
- Test Type: Performance.
- PASSED

2.1.12 Face Recognition - Enroll Users

- This test ensures that the face library can save users.

- Input: User runs the program, enters their name into the terminal and adds themselves as an user.
- Output: Terminal prints “(User’s name) was added successfully
- Test Type: Performance.
- PASSED

2.1.13 Face Recognition - Identify Multiple Users

- This test ensures that the face library can identify multiple users.
- Input: User1 appears into the camera, followed by User2 and User3.
- Output: Shows “User1. User2. User3.”
- Test Type: Performance.
- PASSED

2.1.14 Face Recognition - Recognize Unauthorized Users

- This test ensures that the face library can identify users who have not been registered and are not authorized.
- Input: Unregistered user appears in the camera.
- Output: “Unknown”
- Test Type: Performance.
- PASSED

2.1.15 Face Recognition - Identify More Than 1 Users at the Same Time

- This test ensures that the face library can identify multiple users at the same time.
- Input: User1 appears into the camera with User2.
- Output: Shows both User1 and User2.
- Test Type: Performance.
- PASSED

2.1.16 Touchscreen: Displays OS

- This test ensures that the Raspberry Pi is configured correctly
- Input: Raspberry Pi OS
- Output: Display to touchscreen
- Test Type: Performance
- PASSED

2.1.17 Raspberry Pi: Connect multiple to SSH

- This test ensures that the raspberry pis are able to talk to each other
- Input: Raspberry Pi OS
- Output: Connection to SSH

- Test type: Performance
- FAILED: we are no longer using 2 Pis and we learned that SSH cannot be used to have 2 Pis communicate anyway.

Initial Integration

2.2.1 Camera- Positional Settings

- This test ensures that the camera's motor can turn to different positions
- Inputs: motion sensor detects movement at section B
- Expected output: camera turn to section B
- Test type: performance
- PASSED

2.2.2 Camera-communication

- This test ensures that the camera is able to communicate with raspberry pi to do facial recognition
- Inputs: human face
- Expected output: face identified and communicate the results to the pi
- Test type: performance
- PASSED/FAILED: This test passes when their is stable WiFi but when we connect to a hotspot some of the capabilities are limited.

2.2.3 Touchscreen and user interface-functionality

- This test ensures that the touchscreen is on, and accurately displays the menu
- Inputs: on, and load SD card
- Expected output: our menu
- Test type: performance
- PASSED

2.2.4 Touchscreen and user interface-usability

- This test ensures that the user is able to use the touchscreen and have the touch screen accurately output what is wanted by the user
- Inputs: user presses button A
- Expected output: touchscreen registers the press and outputs the required action
- Test type: usability
- PASSED

2.2.5 Touchscreen and user interface-data storage

- This test ensures that the raspberry pi stores data given by the user

- Inputs: user creates a profile
- Expected output: touchscreen registers the press and stores
- Test type: performance
- PASSED

Final Components

3.1.1. Facial recognition and Voice

- This test verifies that a face can be recognized as a user
- Input: user face and voice
- Output: identification
- Test type: performance
- PASSED/FAILED: When we run it on the hotspot, this feature does not work. It cannot do all of the features on the PI unless there is stable WiFi.

3.1.2 Voice recognition on Pi

- This test verifies that the pi can tell the difference between voices of different users
- Input: 2 different voices
- Output: identification of the 2 separate users
- Test type: performance
- PASSED

3.1.3. Pincode

- This test verifies that the pi can store and identify a user pincode
- Input: user pincode
- Output: identification
- Test type: performance
- PASSED

Final Integration

3.2.1. 2-step authentication to ARM - with voice

- This test verifies that the pi can identify any user with permissions and ARM the system using voice
- Input: face and voice
- Output: signify system ARMED
- Test type: performance

- FAILED: we are no longer using voice to arm the system

3.2.2. 2-step authentication to ARM - with password

- This test verifies that the pi can identify a user with permissions and ARM the system using password
- Input: face and password
- Output: signify system ARMED
- Test type: performance
- PASSED

3.2.3. 3-step authentication to DISARM

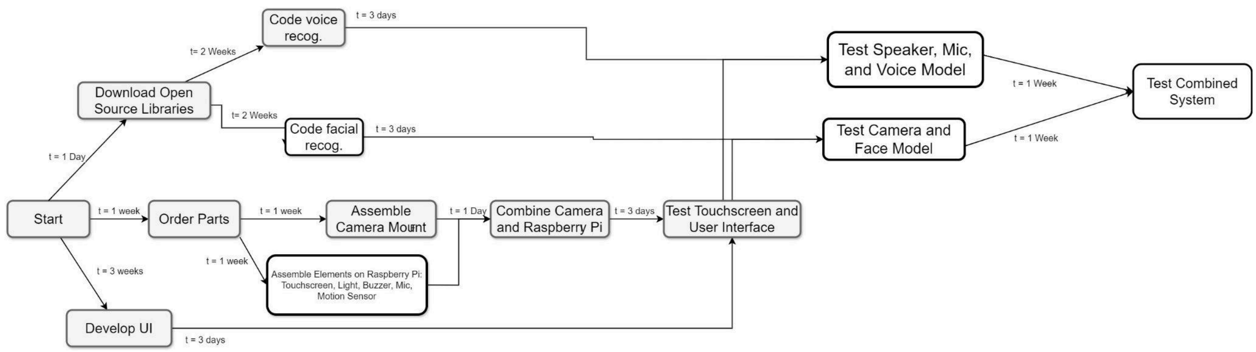
- This test verifies that the pi can identify a user with permissions and ARM the system using password and voice
- Input: face, voice, and password
- Output: signify system DISARMED
- PASSED/FAILED: When we run it on the hotspot, this feature does not work. It cannot do all of the features on the PI unless there is stable WiFi.
    - For the demos we are using motion sensor to set off the alarm and the pincode to disarm

3.2.4. Voice - Arming the System

- This test verifies that GISS can be armed by the admins voice passcode
- Input: Admin speaks the wake-word “Hey Guardian” followed by their voice passcode “Arm”
- Output: LED turns green, “System Armed”
- Test type: Integration
- PASSED

Future Plans

Updated PERT/Gantt Chart



This chart reflects the changes that our project has gone through. Specifically, our major change was removing machine learning before Milestone 1. We successfully finished individual sections before Milestone 2, and just had the integration and testing saved for the end. Along with this, we realized that developing the GUI is a continuous process that cannot have a deadline, so we changed it throughout the development and testing.

Future Directions

Our next steps for the project is reconfiguring it to a stronger wifi, thus enabling the full functionality of the facial and voice recognition. We would want to buy a better camera, as the one we have is not as good quality as we wanted it to be. If this were to be a product in the real world we would want some sort of alert message that could automatically contact the authorities. Aesthetically as well, we would want our product to improve. Right now the menu system, specifically, looks very boring and bland, and we also plan to implement a logo on the casing.

End of Life (EoL) Plan

We have decided to let Briana retain ownership of the GISS as she wanted to do some more work with the fully configured product. We are all willing to continue working on it with her.

Team

Parnika Nammi is a fourth year Computer Science major. She has a passion for embedded systems and machine learning. She worked primarily on the developing logistics for and implementation of the GUI for the GISS. This involved the integration of the face and voice into the main functionality of the process. She also tested the motion sensors to determine the proximity of an individual that can be detected as well as the email notification system for the activity log.

Yongyan Liang is a fourth year Computer Science major. She has a passion for embedded systems and machine learning. She worked primarily on people detection that leads to facial recognition where once motion is detected, people detection will try to detect for a person. She also focused on facial recognition where once a person is detected, it will try to detect their face and check if it is an authorized user. She did user interaction for facial recognition as well that allows deleting a user which removes a user and adding a user which adds a user to the system. She also completed remote viewing for the camera for the user to see the view of the camera offline from anywhere. Additionally, she also did testing for facial recognition and remote viewing to ensure they work before integration with the whole system.

Briana Moyer is a 4th year Computer Engineering student at the University of California, Riverside. While southern California is her home, her love of travel took her all over the world where she eventually found her passion for machine learning and quantum computing. Her primary role in the GISS project was the development of the voice features for GISS and hardware assembly. Coming from a family of builders, she also used her knowledge of power tools and woodwork to build the housing for the display panel and camera module. When asked about her favorite part of the project, she responded with “late nights coding with the girls!”.

Pajaka Lakshmin is a 4th year Computer Science student at the University of California, Riverside. She began coding during elementary school as the lead programmer of her robotics team and has developed a passion for machine learning and embedded systems. Her primary role in the GISS project was the integration of all of the components along with the encryption of the user information. She also worked alongside Briana to test and configure the Raspberry PI.

References

- NEMA WD-7: Occupancy Motion Sensors
- NIST Special Publication 800-63B - Passwords
- Facial Recognition ISO 19794-5
- IEEE Standard for System, Software, and Hardware Verification and Validation (V&V) – IEEE 1012-2016

Acknowledgments

- Aaron
    - He helped us a lot with questions about our pi and the project in general. We were having a lot of connection issues and he would always assist us by providing us with solutions that they tried.

Picture of your team with your final product